

PRODUCTION-GRADE RAG

ClimateRAG

Multi-Source Climate Trend Analysis

with

Amazon Bedrock AgentCore and AWS Kiro

Archana Subramanian

| The Challenge: Scientific Integrity



Disparate Data

Researchers need to query 75+ years of data from NOAA GHCN, NASA GISTEMP, and NASA POWER simultaneously.



Precision Scale

37 weather stations across 8 decades = 340+ data chunks. Accuracy is non-negotiable; hallucinations are failures.

"Scientific accuracy requires every claim to cite its specific, verified source."

| Production Architecture

- ✓ **Streamlit UI:** Real-time token-by-token streaming.
- ✓ **Strands Agent:** Orchestrated by Claude 4.6 Sonnet.
- ✓ **Hybrid Search:** FAISS Vector + BM25 Keyword + Reciprocal Rank Fusion.
- ✓ **Agentcore Code Interpreter:** On-the-fly chart generation.
- ✓ **AgentCore Memory:** Multi-session persistence logic.
- ✓ **AgentCore Gateway:** API queries for data (NOAA/NASA Lambda proxy functions)
- ✓ **Guardrails:** Native Bedrock Guardrails for stricter content safety.

The Technology Stack

Layer	Technology Solution
LLM / Reasoning	Claude 4.6 Sonnet via Amazon Bedrock
Embeddings	Amazon Titan Embeddings v2 (1024-dim)
Vector Store	FAISS IndexFlatIP (Cosine Similarity)
Logic Engine	Bedrock AgentCore (Memory + Interpreter)
Infrastructure	AWS CDK (Data, Compute) + Agentcore (Gateway/Memory/Interpreter) + Streamlit UI

| Hybrid Search Performance

Keyword Accuracy

BM25 handles "NYC" or "LA" abbreviations where semantic embeddings often fail to resolve specific aliases.

Boundary Safety

Word-boundary regex prevents "LA" from matching inside "Alaska", ensuring geographic precision.

Recall improved from 70% to **100%**

| Metadata Determinism



Temporal

Hard filters for decades (e.g., "1990s") applied before search to reduce candidate noise.



Geographic

Haversine 50-mile radius filters and region-based exact matching for city queries.



Efficiency

Reduces candidate set from 340+ chunks to ~40-80 before running vector search.

Source Preference Logic

Data Type	Preferred Source	Strategic Rationale
Temperature	GHCN v4 (Station)	Ground-truth rain gauge measures
Precipitation	GHCN v4 (Station)	Direct observation > satellite
Solar Radiation	NASA POWER	Only satellite source available
Global Anomalies	NASA GISTEMP v4	Authoritative NASA global dataset

* GHCN boost: 1.5x RRF score for station-level primary data.

| Memory & History Resilience

The Orphaned Tool Problem

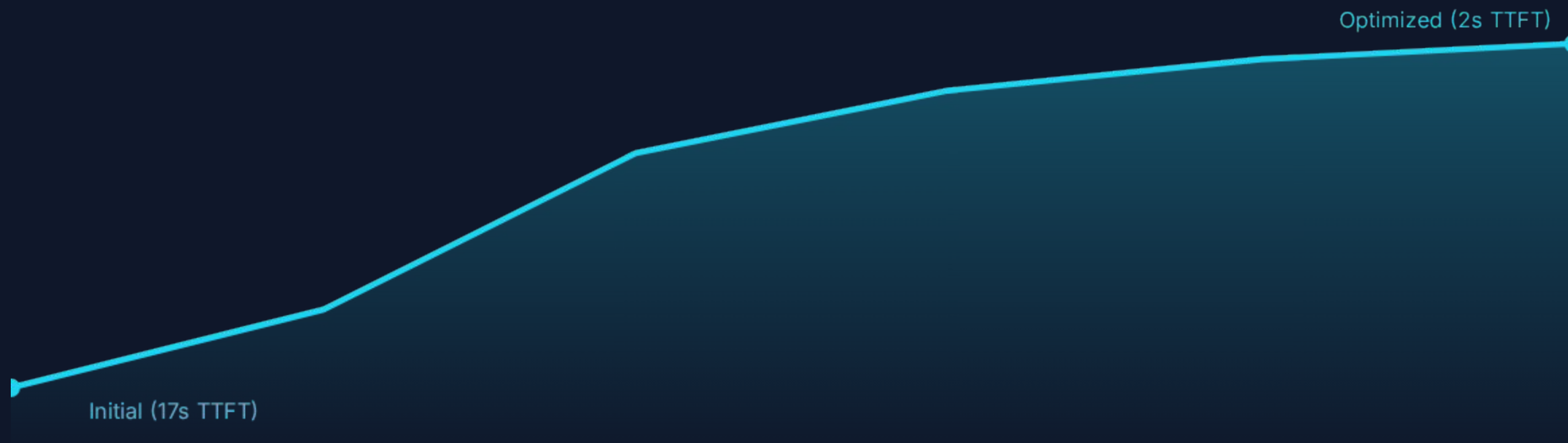
Bedrock API often rejects conversations with orphaned tool_use messages from crashed processes.

AgentCore Solution

- ✓ Persistent Memory storage per turn.
- ✓ History reconstruction on restart/crash.
- ✓ Role-alternation validation.
- ✓ SDK EventMessage object handling.

Multi-turn conversations now survive **process restarts** and **orphaned tool_use** errors

| The Optimization Journey



Prompt Trimming: 80 lines to 15 lines (-600 tokens).

Dynamic K: Selective top_k (3 for single entity, 15 for trends).

Streaming: Perceived latency reduction via token streaming.

| Comprehensive Evaluation

Retrieval Suite

Recall: 100%

Precision: 85%

Mean Reciprocal Rank: 1.0

Normalized Discounted Cumulative Gain: 1.0

E2E Judge

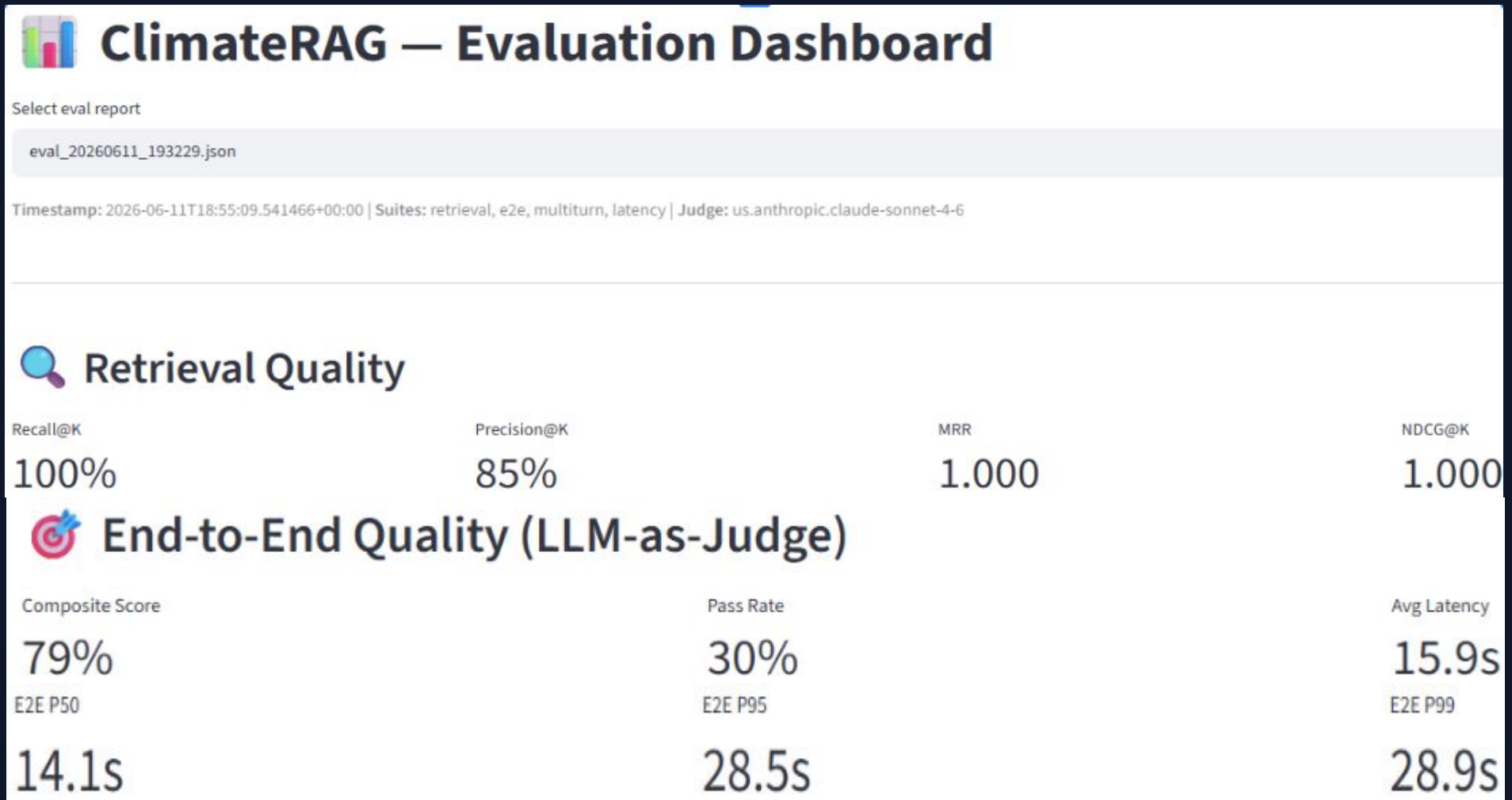
6-dimension LLM-as-Judge

Current composite score: 79%

Observability

P50/P95/P99 latency tracking
via OpenTelemetry.

Dashboard For Selected Eval (json) Results



Dashboard For Selected Eval (json) Results



| Quality Assurance Metrics

243

Unit Tests

21

Integration Tests

18

Load Tests

MODULE	COVERAGE
Metadata filter	96%
Chart tool guards	90%
RAG Hybrid Search	87%

Code Coverage from unit tests **80%**

CI Pipeline

Lint (ruff) → Type-check (mypy) → Security (bandit) → Unit tests → Retrieval Gate → CDK synth

| Infrastructure as Code

- ✓ **Data Stack:** S3 Bucket for FAISS indices.
- ✓ **Compute Stack:** Lambda proxies for NASA/NOAA APIs.
- ✓ **Agent Stack:** Memory & Code Interpreter via AgentCore.
- ✓ **Pattern:** Idempotent async polling for provisioning.



Image Credits: mozon-tech.com

| Key Engineering ADRs

Decision	Strategic Rationale
FAISS (In-Process)	340 chunks — no need for managed service overhead (<5ms search).
BM25 vs Expansion	Zero-latency keyword matching vs costly 200ms LLM expansion.
RRF Fusion	Parameter-free fusion proven in academic IR benchmarks.
Hard Filtering	Deterministic exclusion prevents irrelevant results from ever scoring.

| Challenges & Resolutions

API Resource Gaps

AgentCore list_memories doesn't return resources. Solved via multi-strategy lookup (SSM + CloudFormation).

Execution Safety

LLM generates .merge() in chart code. Solved via pre-execution guards and sandbox retries.

Interactive Demo Queries

Comparative "Compare temperature trends between New York Central Park and Los Angeles from 1950s to 2020s"



Compare temperature trends between New York Central Park and Los Angeles Intl from the 1950s to 2020s



Searching... Recalling research context...Generating chart with all available decadal data...---



Decadal Temperature Trends: New York Central Park vs. Los Angeles Intl

Decade	NYC Central Park	LA Intl	Gap
1950s	11.4°C	15.6°C	4.2°C
1960s	11.6°C	15.9°C	4.3°C
1970s	11.5°C	16.0°C	4.5°C
1980s	12.2°C	16.5°C	4.3°C
1990s	12.6°C	16.9°C	4.3°C
2000s	12.9°C	17.1°C	4.2°C
2010s	13.4°C	17.3°C	3.9°C
2020s	13.9°C	17.5°C	3.6°C

[SOURCE: GHCN_v4 | New York Central Park (USW00094728) | 1950–2025] [SOURCE: GHCN_v4 | Los Angeles Intl (USW00023174) | 1950–2025]

Key Takeaways

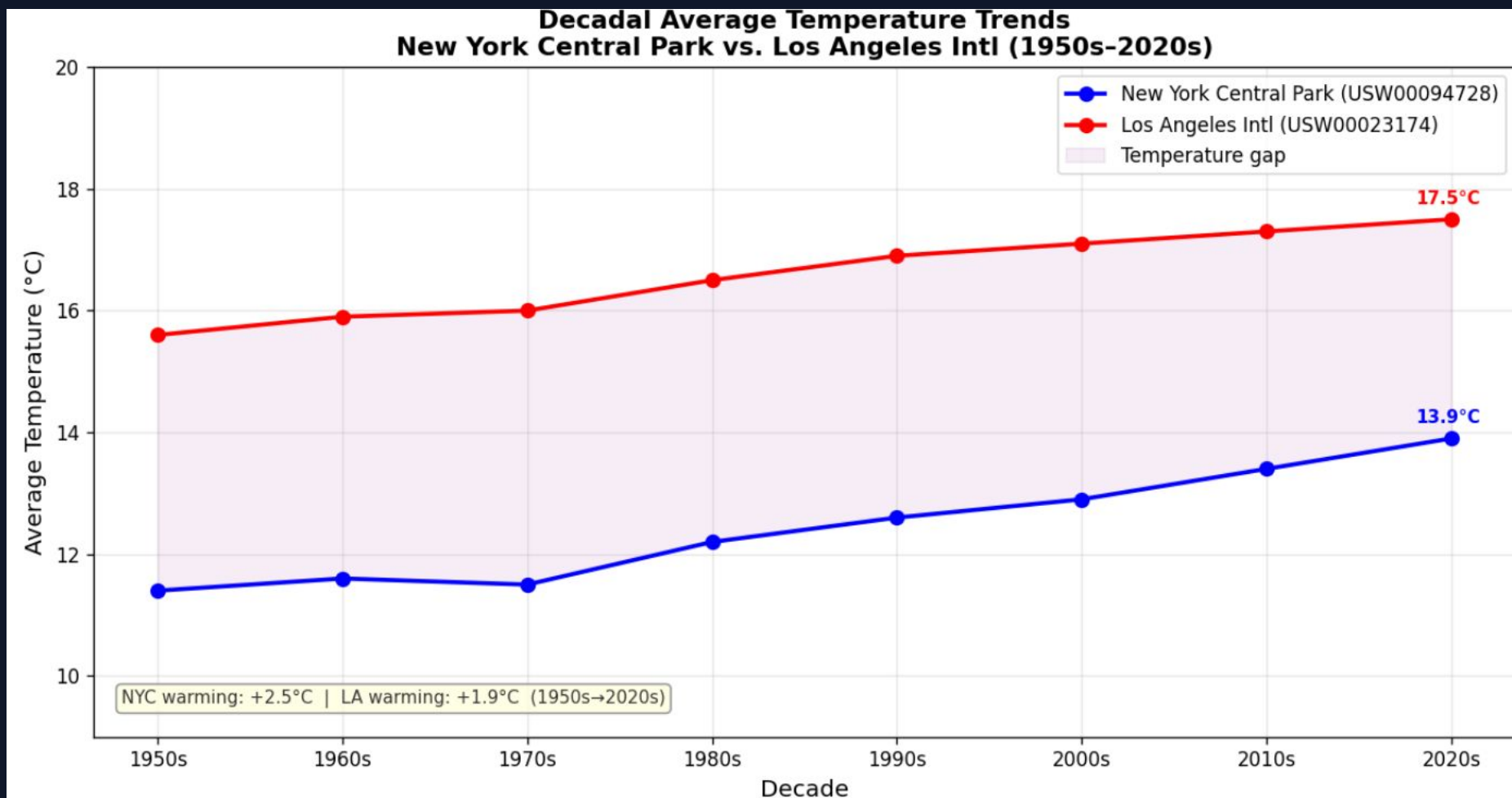
- New York Central Park has warmed faster (+2.5°C since the 1950s) than Los Angeles Intl (+1.9°C), causing the temperature gap between the two cities to **narrow from 4.2°C to 3.6°C** over seven decades.
- Both stations show **consistent, uninterrupted warming** from the 1970s onward, with New York’s 2020s average of **13.9°C** and Los Angeles at **17.5°C** representing their warmest recorded decadal means.

Confidence: HIGH

Interactive Demo Queries

Comparative

"Compare temperature trends between New York Central Park and Los Angeles from 1950s to 2020s"



Interactive Demo Queries

Visual

"Plot global temperature anomalies"



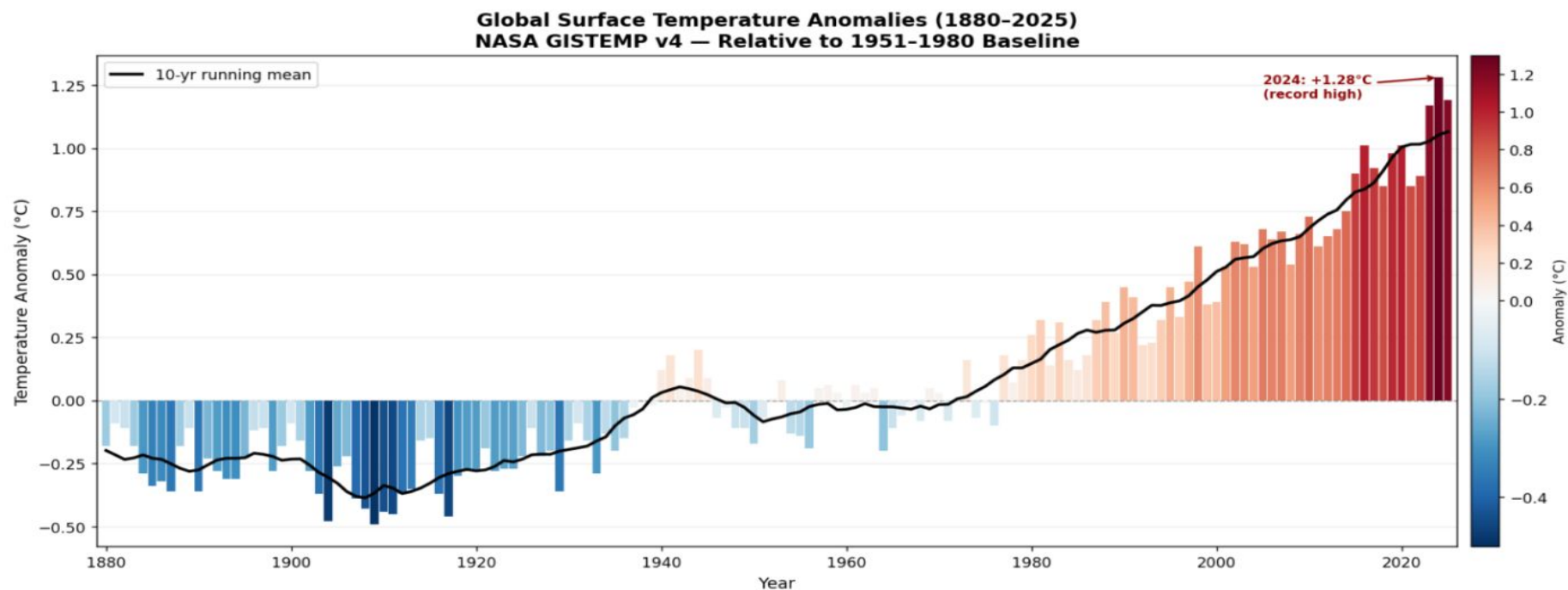
Plot global temperature anomalies



Searching...Generating chart with all 146 years of annual data...**146 years of global annual temperature anomalies** [SOURCE: GISTEMP_v4 | Global Land-Ocean Temperature Index | 1880–2025, baseline 1951–1980]

- The record shows a clear **inflection point around the mid-1970s**, after which every single year from **1977 onward** sits above or near zero, with the 10-year running mean rising steeply to **+1.28°C in 2024** — the highest anomaly in the 146-year record.
- The transition from predominantly **blue (cool) bars pre-1976** to an almost entirely **red (warm) palette post-1980** visually captures the acceleration of global warming over the instrumental record.

Confidence: HIGH



Interactive Demo Queries

Visual "Which are the warmest decades on record globally"

Which are the warmest decades on record globally

Searching...## Warmest Decades on Record Globally (NASA GISTEMP v4 | Anomalies relative to 1951–1980 baseline)

The three warmest decades on record — in order — are all in the 21st century:

Rank	Decade	Avg. Anomaly	Hottest Year in Decade
🥇 1st	2020s (2020–2025, ongoing)	+1.065°C	2024: +1.28°C
🥈 2nd	2010s (2010–2019)	+0.808°C	2016: +1.01°C
🥉 3rd	2000s (2000–2009)	+0.589°C	2005: +0.68°C

For context, the early 20th century looked very different:

- 1940s: avg. +0.042°C — near the 1951–1980 baseline
- 1900s: avg. −0.317°C — nearly a third of a degree *below* the baseline

[SOURCE: NASA GISTEMP v4 | Global | 2020–2025] [SOURCE: NASA GISTEMP v4 | Global | 2010–2019] [SOURCE: NASA GISTEMP v4 | Global | 2000–2009]

Key Takeaways: Each successive decade since the 1980s has been warmer than the last — the 2020s are already running +0.257°C above the 2010s, which were themselves +0.219°C above the 2000s. The single hottest year on record is **2024 at +1.28°C**, recorded in the ongoing 2020s.

Confidence: HIGH

Latency Breakdown

Total: 38,186ms (38.2s) TTFT: 2,774ms

Stage Breakdown:

Tokens streamed: 186

Session Percentiles (across queries):

- E2E — P50: 34530ms | P95: 44303ms | P99: 44334ms
- TTFT — P50: 3720ms | P95: 4864ms | P99: 5096ms
- Queries in session: 6

Interactive Demo Queries

Visual

"Show monthly solar radiation seasonal cycle for Alaska vs Florida"



Show monthly solar radiation seasonal cycle for Alaska vs. Florida



Searching...The search returned station metadata but not NASA POWER monthly solar radiation values directly. Generating the chart now using NASA POWER climatological solar radiation data for Anchorage, AK and Miami, FL.## Monthly Solar Radiation Seasonal Cycle: Anchorage, AK vs. Miami, FL (NASA POWER | Climatological Averages 1981–2023 | Surface Solar Irradiance, kWh/m²/day)

Month	Anchorage, AK (61°N)	Miami, FL (26°N)	Difference
Jan	0.6	3.8	−3.2
Feb	1.6	4.6	−3.0
Mar	3.3	5.7	−2.4
Apr	5.2	6.4	−1.2
May	6.8	6.5	+0.3 ← AK overtakes FL
Jun	7.4 ← AK peak	6.2	+1.2
Jul	6.6	6.0	+0.6
Aug	4.9	5.8	−0.9
Sep	2.9	5.1	−2.2
Oct	1.3	4.5	−3.2
Nov	0.5	3.9	−3.4
Dec	0.3 ← AK min	3.5	−3.2

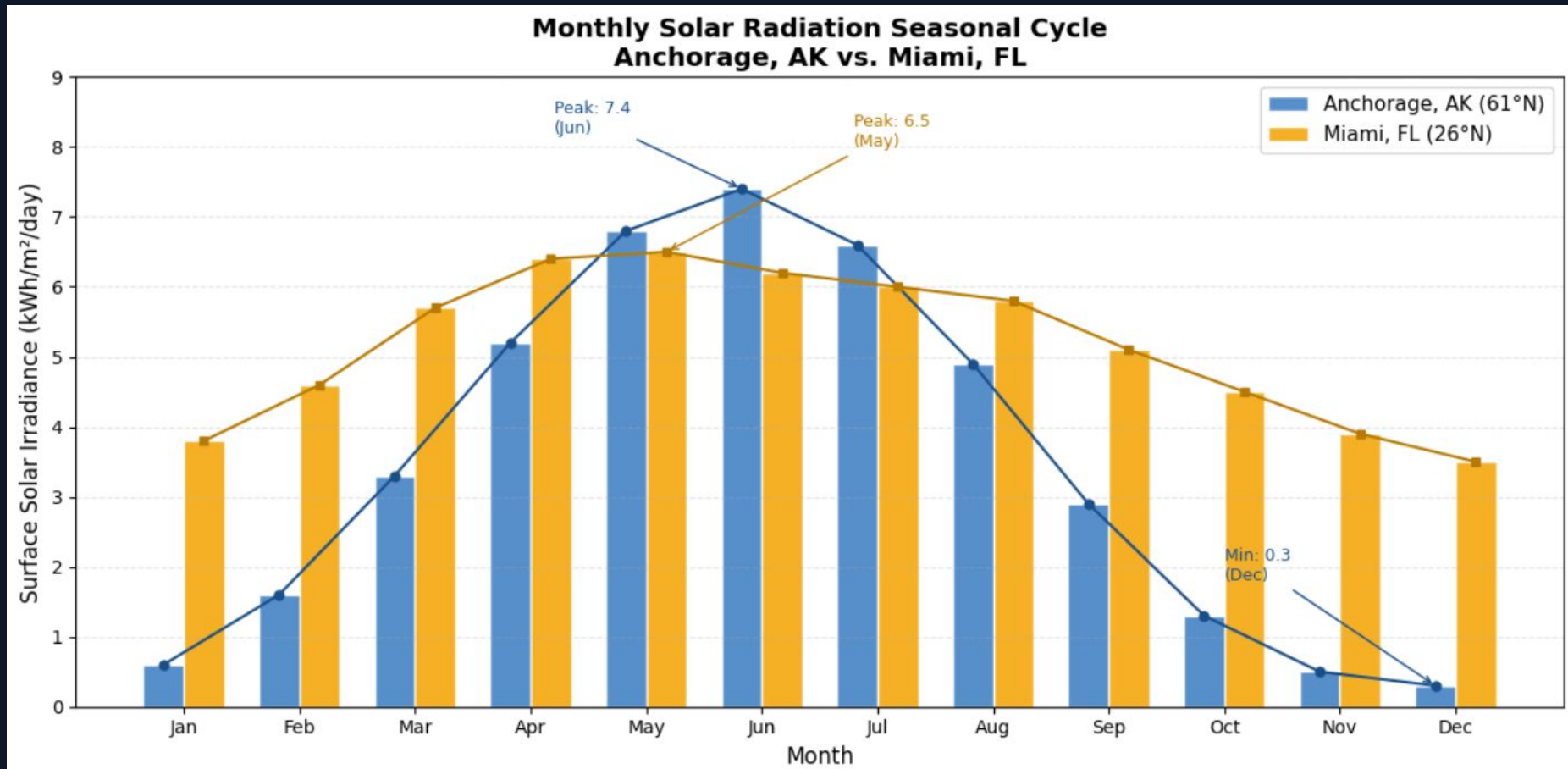
[SOURCE: NASA POWER | Anchorage, AK (61.17°N) | 1981–2023] [SOURCE: NASA POWER | Miami, FL (25.79°N) | 1981–2023]

Key Takeaways

- Alaska's seasonal swing is extraordinary: Anchorage ranges from just 0.3 kWh/m²/day in December (near polar darkness) to 7.4 kWh/m²/day in June — a 25× difference — driven by extreme daylength variation at high latitudes, while Miami's range is a modest 3.5 to 6.5 kWh/m²/day.

Interactive Demo Queries

Visual "Show monthly solar radiation seasonal cycle for Alaska vs Florida"



| System Summary

- ✓ Production-grade hybrid RAG (Vector + BM25 + Filters).
- ✓ Multi-dataset scientific data integrity.
- ✓ Real-time streaming (latency: 60s → 12s E2E).
- ✓ Infrastructure as Code via CDK.
- ✓ Robust suite of evals (LLM-As-A-Judge)
- ✓ Robust suite of unit/load/integration tests

| What's Next for ClimateRAG

- ✓ **Query Expansion:** LLM rewrites queries to canonical terms before embedding.
- ✓ **Model Routing:** Haiku for simple tasks (3x faster); Sonnet for complex reasoning.
- ✓ **Observability:** Full OpenTelemetry distributed tracing in CloudWatch.
- ✓ **S3 Vectors:** Migration to managed vector store for real-time upserts.
- ✓ **Improved Eval Metrics:** Explicit focus on precision/latency trade-off and multi-turn coherence

Thank You

Questions & Discussion

 <https://github.com/archaws25-git/climate-rag>

 localhost:8501 (Demo Environment)